Propagated class exclusion for direct semi-supervised non-negative matrix factorization

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Abstract

Semi-supervised non-negative matrix factorization (NMF) often uses labels to learn a representation before making a separate class decision. We propose propagated class-exclusion NMF (PCE-NMF), which assigns a class meaning to each factor component and uses graph-derived harmonic evidence to penalize unsupported class contributions. Exact class support is imposed for labeled samples, while unlabeled samples are assigned directly from scale-invariant contribution energies. We evaluate PCE-NMF on ten data sets using 20 paired splits, eight established methods, and two mechanism controls. PCE-NMF outperforms a matched hard-exclusion factorization on all ten data sets, demonstrating the effect of incorporating propagated evidence into the factorization. It achieves the highest accuracy among PCE-NMF, HLPNMF, and GOCNMF on COIL20, ORL, and COIL100. These results characterize a direct class-aware factorization in which the same class contributions guide optimization and prediction.

Keywords: Non-negative matrix factorization, Semi-supervised clustering, Label propagation, Class exclusion, Direct clustering

1. Introduction

Non-negative matrix factorization represents a non-negative data matrix through a small set of non-negative factors [1, 2]. In clustering, the sample factor indicates how strongly each sample is associated with each component.

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NMF has been used for document clustering [3]; its connections to spectral and orthogonal clustering formulations have also been studied [4, 5]. Sparsity constraints [6] and alternative numerical solvers [7, 8] address different aspects of the factorization. Limited class labels can make components easier to interpret: they may shape a representation [9], constrain labeled coordinates, or propagate class information to unlabeled samples [10].

The class meaning of an NMF component is often assigned after factorization. With a small labeled set, it can instead be specified during fitting: each of the c components denotes one class, and the labeled rows of the sample factor have exact class support. This constraint identifies the labeled components but does not determine which class contributions should be favored at unlabeled samples. A harmonic field on the sample graph supplies class evidence at those samples.

PCE-NMF uses the complement of that evidence as a class-specific penalty on each factor contribution. The same contribution energy determines the final class decision. Graph information therefore acts on candidate class contributions during fitting, while the data term retains ordinary NMF reconstruction. The contribution score is also invariant to positive diagonal rescaling of the two factors.

We test whether propagated evidence improves a factorization with hard labeled support, whether the factor-based decision changes the predictions of the underlying harmonic field, and how PCE-NMF compares with established semi-supervised NMF methods under matched splits and their native prediction rules. Ten-data-set experiments address these questions. Prespecified analyses examine label fraction, corruption, class count, runtime, and numerical convergence. The controls show a consistent improvement over hard exclusion; the comparison with the harmonic predictor identifies where factor-based decisions gain or lose accuracy.

The contribution is a class-semantic NMF formulation that applies harmonic evidence to the energy of each candidate class contribution and makes its decision from that same energy. We establish well-posedness of the regularized harmonic step, descent of the exact alternating updates under stated conditions, and scale invariance of the class score. Matched controls distinguish the effect of propagated exclusion during fitting from the predictive contribution of the harmonic field. The nine-method comparison measures the resulting accuracy profile and computational cost.

2. Related Work

2.1. Representation-level semi-supervision in NMF

Constrained NMF incorporates partial labels into the factorization to learn a discriminative representation [9]. Other representation-level methods use discriminative terms on labeled samples [11, 12], local constraints [13], or a combination of label discrimination and graph regularization [14]. The learned coordinates can then support a downstream class decision. PCE-NMF uses c components with explicit class meanings and restricts the labeled rows of V to their known class support. Its unlabeled penalty acts on the contribution of each class component to each sample, so the class decision is obtained from the factors themselves.

2.2. Graph and propagation mechanisms

Graph-based clustering includes normalized cuts [15], Laplacian embeddings [16], and spectral clustering [17]. Graph-regularized NMF preserves local relations in the low-dimensional representation [18]; symmetric NMF provides another factorization of a sample affinity matrix [19]. Harmonic label propagation solves a graph boundary-value problem to extend labels from observed to unobserved vertices [10], while local and global consistency uses a graph-smoothing formulation [20].

In semi-supervised NMF, pairwise constraints can modify the graph [21]. Label and constraint propagation have also been combined in the factorization [22]; hypergraph methods use higher-order sample relations [23], and pivotal-aware models propagate both point labels and pairwise constraints [24]. Weighted label propagation is coupled to NMF in SNMFWLP [25]. HLPNMF couples high-order propagation with NMF through a hypergraph-induced transition structure [26]. PCE-NMF computes a harmonic class field first and uses it as fixed class-exclusion evidence during factorization. Thus its graph affects the class penalties, while the reconstruction remains the ordinary two-factor NMF term.

2.3. Direct class-semantic factorization

GOCNMF embeds one-hot labels in the representation of labeled samples and applies graph regularization to guide unlabeled representations [27]. PCE-NMF also assigns class meanings to factor coordinates, but it uses hard class support rather than fixing labeled coordinates to one-hot values. For unlabeled samples, it penalizes class-specific reconstruction energy according to harmonic

evidence. This separates the observed label constraint from the propagated preference over candidate classes. Robust semi-supervised NMF [28] and self-supervised semi-supervised NMF [29] pursue complementary goals through robust losses or additional self-supervisory signals.

Table 1 compares the supervision and decision mechanisms of the closest evaluated methods. GOCNMF anchors class coordinates and regularizes the representation through a graph; SNMFWLP and HLPNMF couple propagation to factor learning in different ways. PCE-NMF weights the reconstruction energy of each class-specific factor contribution by a fixed harmonic field and uses that energy for prediction.

Table 1: How closely related methods use labels and make class decisions. The entries describe the formulations and native prediction rules used in the paired comparison.

Method	Role of supervision in the factorization	Native decision
GOCNMF	One-hot labeled coordinates and graph regularization of the representation	Maximum representation coordinate
SNMFWLP	Weighted label propagation coupled to NMF with basis orthogonality	K-means on the representation
HLPNMF	High-order propagation regularization through a shared coefficient matrix	Maximum class-semantic coordinate
PCE-NMF	Harmonic evidence weights class-specific contribution energy under exact labeled support	Maximum contribution energy

3. Propagated Class-Exclusion NMF

3.1. Problem formulation

Let $X = [x_1, \ldots, x_n] \in \mathbb{R}_+^{d\times n}$ contain n non-negative samples with d features, and let c be the number of classes. We factorize $X \approx UV^\top$ with $U = [u_1, \ldots, u_c] \in \mathbb{R}_+^{d\times c}$ and $V = (v_{ik}) \in \mathbb{R}_+^{n\times c}$. A labeled index set $\mathcal{L} \subseteq \{1, \ldots, n\}$ supplies labels $y_i \in \{1, \ldots, c\}$ for $i \in \mathcal{L}$. The class identity of component k is fixed by these labels. Nonzero samples are normalized to unit Euclidean norm before graph construction and factorization.

3.2. Harmonic class evidence

Construct a binary p-nearest-neighbor graph from Euclidean distances between the normalized samples and symmetrize its adjacency matrix $W \in$

$\{0,1\}^{n\times n}$. Let $D = \mathrm{diag}(W\mathbf{1})$ and $L = D - W$. For each connected component containing both labeled and unlabeled vertices, partition its Laplacian by the labeled and unlabeled indices. The labeled one-hot matrix C defines the regularized harmonic field

$$(L_{\mathcal{U}\mathcal{U}} + \varepsilon I)F_{\mathcal{U}} = W_{\mathcal{U}\mathcal{L}}C, \qquad \varepsilon = 10^{-12}. \tag{1}$$

The implementation clips negative numerical entries to zero and normalizes each nonzero row to sum to one. A component without labeled vertices contributes no propagated exclusion. Let $P \in [0,1]^{n\times c}$ denote the resulting class evidence on anchored unlabeled vertices; the known one-hot labels are retained on $\mathcal{L}$.

3.3. Propagated exclusion model

Define the exclusion weights $Q \in [0,1]^{n\times c}$ by

$$Q_{ik} = \begin{cases} 0, & i \in \mathcal{L},\ k = y_i, \\ 1, & i \in \mathcal{L},\ k \neq y_i, \\ 1 - P_{ik}, & i \notin \mathcal{L} \text{ in an anchored component}, \\ 0, & i \notin \mathcal{L} \text{ in an unanchored component}. \end{cases} \tag{2}$$

The PCE-NMF objective is

$$\min_{U,V\geq 0} \quad \mathcal{J}(U,V) = \|X - UV^\top\|_F^2 + \lambda \sum_{i=1}^{n}\sum_{k=1}^{c} Q_{ik}\|u_k\|_2^2 v_{ik}^2, \tag{3}$$

Here $\lambda \geq 0$ controls exclusion strength. The weight $1 - P_{ik}$ penalizes a class contribution more strongly when the graph provides less evidence for that class. The hard labeled-support constraint $v_{ik} = 0$ for $i \in \mathcal{L}$ and $k \neq y_i$ remains exact throughout optimization.

3.4. Direct clustering rule

After optimization, sample i receives the class

$$\hat{y}_i = \arg\max_{1\leq k\leq c} s_{ik}, \qquad s_{ik} = \|u_k\|_2^2 v_{ik}^2. \tag{4}$$

This rule uses the class meaning of the factors directly and assigns samples from V in one step. Figure 1 summarizes the path from labeled samples through graph-derived exclusion to the class decision.

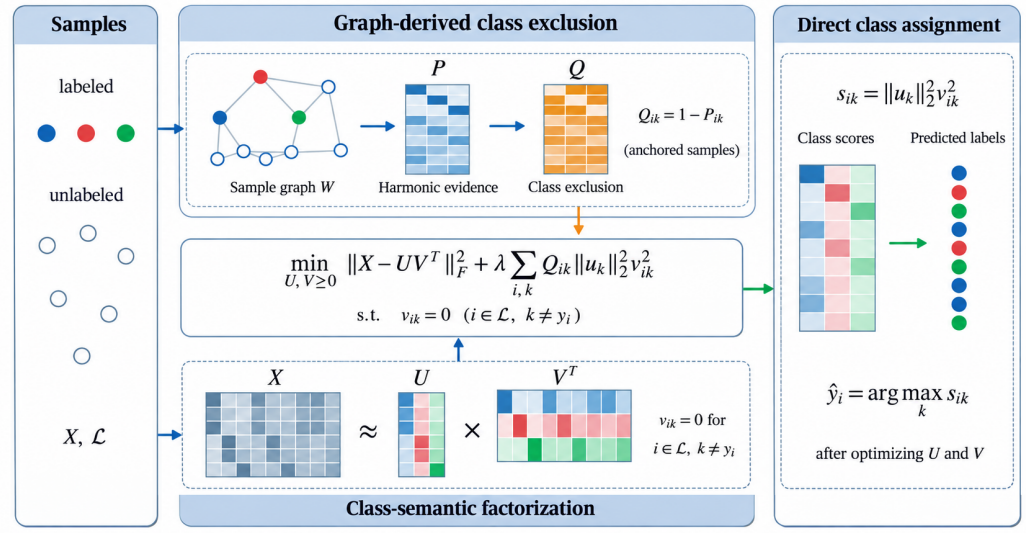


Figure 1: PCE-NMF mechanism. Labeled samples anchor harmonic class evidence P on the sample graph; the resulting exclusion weights Q guide class-semantic factorization. After optimization, contribution energies give direct class assignments. The heatmaps and sample colors are schematic.

3.5. Optimization

Write $a_k = \|u_k\|_2^2$ and $b_k = \sum_i Q_{ik} v_{ik}^2$. The gradients of the objective are

$$\nabla_U \mathcal{J} = 2\Big(UV^\top V - XV + \lambda U \operatorname{diag}(b)\Big), \tag{5}$$

$$\nabla_V \mathcal{J} = 2\Big(VU^\top U - X^\top U + \lambda (Q \odot V) \operatorname{diag}(a)\Big). \tag{6}$$

The implemented alternating multiplicative updates are

$$U \leftarrow U \odot \frac{XV}{UV^\top V + \lambda U \operatorname{diag}(b)}, \tag{7}$$

$$V \leftarrow V \odot \frac{X^\top U}{VU^\top U + \lambda (Q \odot V) \operatorname{diag}(a)}. \tag{8}$$

The fractions are elementwise. Denominators are bounded below by 10^{-12} in the MATLAB implementation. After each update, forbidden labeled coordinates are reset to zero. The formal protocol uses 50 alternating updates and the parameters locked before evaluation.

Algorithm 1: PCE-NMF optimization..

1. **Input:** Non-negative data X, labeled indices $\mathcal{L}$ and labels y_i, class count c, graph size p, locked penalty λ, and seed. Set $T = 50$.
2. Normalize nonzero columns of X; construct the symmetric binary p-nearest-neighbor graph W. In each anchored component solve Eq. (1), normalize its nonzero rows, and form Q by Eq. (2).
3. Initialize non-negative U and V from the recorded seed. Set $v_{ik} = 0$ for labeled i and $k \neq y_i$.
4. For $t = 1, \ldots, T$, update U and then V by Eqs. (7)–(8); bound denominators below by 10^{-12} and reset forbidden labeled entries of V to zero.
5. **Output:** Factors U, V and unlabeled assignments from Eq. (4).

3.6. Computational complexity

The blocked distance calculation for the graph requires $O(dn^2)$ arithmetic; selecting neighbors by sorting each distance column adds $O(n^2 \log n)$ comparisons. The symmetric binary graph contains $O(np)$ edges. The harmonic step solves one sparse system per anchored connected component; its time and memory depend on sparse factorization fill-in. Each factor update costs $O(dnc + (d+n)c^2)$ arithmetic and stores $O((d+n)c)$ factor entries, in addition to X and the graph. These bounds describe the implementation's main operations; measured wall times are reported in Section 5.

4. Theoretical Analysis

The descent proof uses the auxiliary-function approach for multiplicative NMF updates [2, 26] and accounts for the class-exclusion terms in both factor blocks.

Proposition 1 (Well-posed regularized harmonic step)**.** *For an undirected graph with non-negative weights and $\varepsilon > 0$, the matrix $L_{\mathcal{U}\mathcal{U}} + \varepsilon I$ in Eq.* (1) *is symmetric positive definite. Hence the regularized harmonic field is uniquely defined for every right-hand side.*

Proof. The graph Laplacian is symmetric positive semidefinite because

$$z^\top L z = \tfrac{1}{2} \sum_{i,j} W_{ij}(z_i - z_j)^2 \geq 0$$

for every $z \in \mathbb{R}^n$. A principal submatrix of a positive semidefinite matrix is positive semidefinite. Therefore, for nonzero $h \in \mathbb{R}^{|\mathcal{U}|}$, $h^\top (L_{\mathcal{U}\mathcal{U}} + \varepsilon I)h \geq \varepsilon \|h\|_2^2 > 0$. Symmetric positive definiteness gives invertibility and a unique solution. □

Theorem 1 (Descent of the exact alternating updates)**.** *Fix $Q \in [0,1]^{n \times c}$, let $X \in \mathbb{R}_+^{d \times n}$, and take $\lambda \geq 0$. Apply Eqs.* (7)–(8) *in exact arithmetic, keeping forbidden labeled coordinates at zero. At each update, assume every positive, permitted coordinate has a positive denominator; any coordinate that reaches zero remains zero. Each factor update then does not increase $\mathcal{J}$. Consequently, the exact alternating sequence has nonincreasing objective values that converge to a finite limit.*

Proof. For fixed V, let $u \in \mathbb{R}_+^c$ be one row of U and x the corresponding row of X. Up to a constant, its objective is $f(u) = u^\top A u - 2h^\top u$, where $A = V^\top V + \lambda \operatorname{diag}(b)$ and $h = xV$. For fixed U, the corresponding expression for row v_i of V has $A_i = U^\top U + \lambda \operatorname{diag}(Q_{i:} \odot a)$ and $h_i = x_i^\top U$, where x_i is column i of X. Both versions of A are symmetric and entrywise nonnegative, and their corresponding h are nonnegative. Restrict a row to its positive, permitted coordinates, write the current vector as $z > 0$, and set $D = \operatorname{diag}((Az)_k / z_k)$. For any δ,

$$\delta^\top (D - A)\delta = \tfrac{1}{2} \sum_{j,k} A_{jk} z_j z_k \left(\frac{\delta_j}{z_j} - \frac{\delta_k}{z_k} \right)^2 \geq 0.$$

Hence $g(z + \delta \mid z) = f(z) + 2(Az - h)^\top \delta + \delta^\top D \delta$ majorizes $f(z + \delta)$ and is equal to it at $\delta = 0$. The positive diagonal matrix D gives the nonnegative minimizer $z^+ = D^{-1}h = z \odot h/(Az)$, which is the multiplicative update on these coordinates. Thus $f(z^+) \leq g(z^+ \mid z) \leq g(z \mid z) = f(z)$. If a numerator vanishes, the updated coordinate becomes zero and is held there; the same argument applies to the remaining active coordinates. Summing over rows proves descent for each factor block. Finally, $\mathcal{J} \geq 0$, so its sequence of objective values converges to a finite limit. □

Theorem 1 establishes convergence of objective values for the exact updates. The MATLAB implementation bounds denominators below by 10^{-12} and clips factor entries to that level; its objective trajectories are measured in Section 5.

Proposition 2 (Scale invariance of class energy)**.** *For any positive diagonal matrix* $T = \mathrm{diag}(t_1, \ldots, t_c)$*, the feasible transformation* $U \mapsto UT$ *and* $V \mapsto VT^{-1}$ *leaves the reconstruction, the exclusion penalty, and every class score in Eq.* (4) *unchanged.*

Proof. Since T is diagonal and positive, $(UT)(VT^{-1})^\top = UV^\top$. For each class k, $\|t_k u_k\|_2^2 (v_{ik}/t_k)^2 = \|u_k\|_2^2 v_{ik}^2$. This identity applies to each term of the exclusion penalty and to each class score. Multiplication by a positive diagonal matrix also preserves non-negativity and every forbidden zero in V. □

For an anchored unlabeled sample, $Q_{ik} = 1 - P_{ik}$ gives the exact identity

$$\sum_k Q_{ik} s_{ik} = \sum_k s_{ik} - \sum_k P_{ik} s_{ik}. \tag{9}$$

The penalty therefore reduces the cost of contributions aligned with the propagated evidence, relative to contributions to unsupported classes. Equation (9) is an algebraic interpretation of the fixed objective; the measured effect of this mechanism is tested by the matched controls in Section 5.

5. Experiments

5.1. Experimental settings

We evaluated NMF, CNMF, GNMFLD, EDNMF, LCNMF, GOCNMF, SNMFWLP, HLPNMF, and PCE-NMF on YaleB, COIL20, Optdigits, ORL, PIE, COIL100, AR, MSRA25, MNIST, and USPS. COIL20 and COIL100 originate from the Columbia Object Image Library [30, 31]; MNIST is the handwritten-digit benchmark described by LeCun et al. [32]. These ten data sets were fixed before the formal comparison and cover face, object, and digit recognition tasks. Nonzero samples were normalized to unit Euclidean norm. For each class, $\max\{1, \lfloor 0.1 n_k \rfloor\}$ samples were labeled, where n_k is the class size. All methods received the same labeled split in each of 20 paired runs. Accuracy (ACC), normalized mutual information (NMI), purity, F-score, precision, and recall were measured on unlabeled samples. The external methods retained their native training and prediction rules. Parameters were fixed before the formal runs using source settings where available and separate development splits for PCE-NMF.

Table 2 gives the actual matrix dimensions and labeled sample counts used in the formal comparison. The number of labeled samples follows the per-class floor rule above; all reported scores use the remaining samples.

Table 2: Data sets in the locked formal comparison. n is the sample count, d is the feature dimension, c is the class count, and $n_{\mathcal{L}}$ is the number of labeled samples per paired split.

Data set	n	d	c	$n_{\mathcal{L}}$
YaleB	2414	1024	38	227
COIL20	1440	1024	20	140
Optdigits	5620	64	10	557
ORL	400	4096	40	40
PIE	2856	1024	68	272
COIL100	7200	1024	100	700
AR	1680	2000	120	120
MSRA25	1799	256	12	173
MNIST	6996	784	10	695
USPS	9298	256	10	925

5.2. Clustering performance

Table 3 summarizes the 20-seed means for PCE-NMF and two graph-based competitors. Tables 4– 13 immediately below report the complete nine-method, six-metric means and standard deviations for each data set. PCE-NMF achieved the highest ACC among these three methods on COIL20, ORL, and COIL100. HLPNMF led on YaleB, Optdigits, PIE, AR, MSRA25, and USPS, while GOCNMF led on MNIST. The pattern changes with the data set and metric: for example, PCE-NMF attained ACC/NMI of 0.9100/0.9452 on COIL100, whereas HLPNMF attained 0.8740/0.9229; on PIE the corresponding values were 0.9376/0.9527 and 0.9589/0.9714.

Figure 2 displays the mean ACC and NMI differences between PCE-NMF and each baseline across the ten data sets.

Table 3: Mean ACC and NMI over 20 paired seeds on unlabeled samples. Full nine-method results follow in this section.

	ACC			NMI		
Data set	PCE-NMF	HLPNMF	GOCNMF	PCE-NMF	HLPNMF	GOCNMF
YaleB	0.6934	0.7308	0.7088	0.6979	0.7323	0.7054
COIL20	0.9713	0.9515	0.9100	0.9725	0.9518	0.9032
Optdigits	0.9832	0.9859	0.9840	0.9579	0.9632	0.9593
ORL	0.6506	0.5935	0.6426	0.7910	0.7904	0.7963
PIE	0.9376	0.9589	0.9352	0.9527	0.9714	0.9534
COIL100	0.9100	0.8740	0.8166	0.9452	0.9229	0.8731
AR	0.6332	0.6390	0.6323	0.8381	0.8418	0.8394
MSRA25	0.9788	0.9941	0.9748	0.9607	0.9878	0.9588
MNIST	0.9315	0.9291	0.9352	0.8501	0.8508	0.8571
USPS	0.9700	0.9719	0.9695	0.9226	0.9268	0.9222

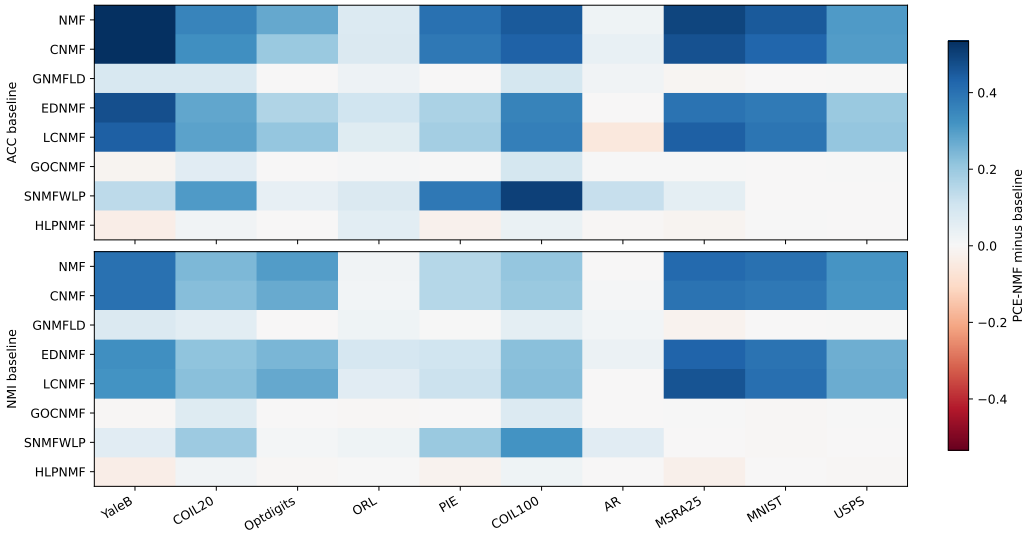


Figure 2: Differences in mean ACC and NMI between PCE-NMF and each baseline on the ten data sets. Each cell averages 20 matched seeds; positive values favor PCE-NMF.

Table 4: Complete formal results on YaleB. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.1611 ± 0.0095	0.2993 ± 0.0143	0.1680 ± 0.0094	0.1023 ± 0.0066	0.0927 ± 0.0068	0.1142 ± 0.0069
CNMF	0.1594 ± 0.0080	0.2984 ± 0.0102	0.1670 ± 0.0087	0.1017 ± 0.0060	0.0915 ± 0.0062	0.1147 ± 0.0066
GNMFLD	0.6091 ± 0.0144	0.6203 ± 0.0105	0.6096 ± 0.0141	0.4087 ± 0.0159	0.3938 ± 0.0180	0.4250 ± 0.0143
EDNMF	0.2227 ± 0.0156	0.3717 ± 0.0125	0.2329 ± 0.0159	0.1411 ± 0.0142	0.1236 ± 0.0170	0.1674 ± 0.0117
LCNMF	0.2567 ± 0.0105	0.3807 ± 0.0075	0.2655 ± 0.0109	0.1463 ± 0.0057	0.1373 ± 0.0057	0.1565 ± 0.0060
GOCNMF	0.7088 ± 0.0131	0.7054 ± 0.0087	0.7089 ± 0.0129	0.5309 ± 0.0138	0.5173 ± 0.0142	0.5452 ± 0.0141
SNMFWLP	0.5544 ± 0.0210	0.6368 ± 0.0185	0.5689 ± 0.0202	0.2177 ± 0.0292	0.1439 ± 0.0271	0.4624 ± 0.0301
HLPNMF	0.7308 ± 0.0129	0.7323 ± 0.0095	0.7308 ± 0.0128	0.5651 ± 0.0155	0.5506 ± 0.0163	0.5804 ± 0.0151
PCE_NMF	0.6934 ± 0.0187	0.6979 ± 0.0142	0.6934 ± 0.0187	0.4952 ± 0.0255	0.4672 ± 0.0295	0.5271 ± 0.0212

Table 5: Complete formal results on COIL20. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.6182 ± 0.0366	0.7323 ± 0.0159	0.6402 ± 0.0312	0.5606 ± 0.0281	0.5235 ± 0.0360	0.6042 ± 0.0234
CNMF	0.6419 ± 0.0327	0.7457 ± 0.0168	0.6614 ± 0.0272	0.5866 ± 0.0269	0.5562 ± 0.0313	0.6211 ± 0.0264
GNMFLD	0.8873 ± 0.0075	0.9146 ± 0.0076	0.8880 ± 0.0079	0.8478 ± 0.0105	0.8356 ± 0.0137	0.8605 ± 0.0115
EDNMF	0.6947 ± 0.0096	0.7594 ± 0.0096	0.6979 ± 0.0100	0.6178 ± 0.0115	0.6013 ± 0.0118	0.6354 ± 0.0119
LCNMF	0.6859 ± 0.0180	0.7492 ± 0.0137	0.6897 ± 0.0154	0.6059 ± 0.0186	0.5887 ± 0.0210	0.6242 ± 0.0163
GOCNMF	0.9100 ± 0.0198	0.9032 ± 0.0225	0.9100 ± 0.0198	0.8389 ± 0.0332	0.8281 ± 0.0355	0.8501 ± 0.0314
SNMFWLP	0.6682 ± 0.0379	0.7803 ± 0.0379	0.7031 ± 0.0361	0.5352 ± 0.0525	0.4531 ± 0.0546	0.6563 ± 0.0543
HLPNMF	0.9515 ± 0.0099	0.9518 ± 0.0093	0.9515 ± 0.0099	0.9110 ± 0.0174	0.9027 ± 0.0197	0.9196 ± 0.0155
PCE_NMF	0.9713 ± 0.0050	0.9725 ± 0.0040	0.9713 ± 0.0050	0.9473 ± 0.0089	0.9411 ± 0.0120	0.9536 ± 0.0066

Table 6: Complete formal results on Optdigits. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.7106 ± 0.0372	0.6578 ± 0.0213	0.7281 ± 0.0314	0.6090 ± 0.0336	0.5894 ± 0.0399	0.6304 ± 0.0301
CNMF	0.7841 ± 0.0451	0.6888 ± 0.0256	0.7893 ± 0.0373	0.6648 ± 0.0370	0.6561 ± 0.0415	0.6741 ± 0.0338
GNMFLD	0.9838 ± 0.0018	0.9586 ± 0.0038	0.9838 ± 0.0018	0.9680 ± 0.0035	0.9678 ± 0.0036	0.9682 ± 0.0035
EDNMF	0.8230 ± 0.0075	0.7131 ± 0.0069	0.8230 ± 0.0075	0.7005 ± 0.0109	0.6955 ± 0.0130	0.7056 ± 0.0093
LCNMF	0.7751 ± 0.0572	0.6865 ± 0.0367	0.7811 ± 0.0544	0.6641 ± 0.0513	0.6573 ± 0.0538	0.6710 ± 0.0492
GOCNMF	0.9840 ± 0.0016	0.9593 ± 0.0033	0.9840 ± 0.0016	0.9684 ± 0.0030	0.9682 ± 0.0031	0.9686 ± 0.0030
SNMFWLP	0.9401 ± 0.0551	0.9455 ± 0.0165	0.9459 ± 0.0474	0.9304 ± 0.0469	0.9044 ± 0.0791	0.9610 ± 0.0132
HLPNMF	0.9859 ± 0.0015	0.9632 ± 0.0031	0.9859 ± 0.0015	0.9720 ± 0.0028	0.9720 ± 0.0029	0.9721 ± 0.0028
PCE_NMF	0.9832 ± 0.0016	0.9579 ± 0.0033	0.9832 ± 0.0016	0.9667 ± 0.0031	0.9665 ± 0.0031	0.9669 ± 0.0030

5.3. Paired statistical analysis

For each data set, we first averaged each method over the same 20 seeds. Across-data-set comparisons then used the ten data-set means as paired units, following the distinction between repeated trials and independent data sets in multi-algorithm comparisons [33]. We used the Friedman rank test [34] for

Table 7: Complete formal results on ORL. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.5769 ± 0.0253	0.7709 ± 0.0146	0.6251 ± 0.0216	0.4296 ± 0.0269	0.3700 ± 0.0294	0.5134 ± 0.0279
CNMF	0.5747 ± 0.0337	0.7743 ± 0.0150	0.6264 ± 0.0263	0.4346 ± 0.0343	0.3730 ± 0.0404	0.5224 ± 0.0259
GNMFLD	0.6235 ± 0.0195	0.7686 ± 0.0128	0.6369 ± 0.0187	0.4438 ± 0.0325	0.3771 ± 0.0410	0.5421 ± 0.0195
EDNMF	0.5461 ± 0.0303	0.7010 ± 0.0214	0.5568 ± 0.0311	0.3554 ± 0.0337	0.3048 ± 0.0359	0.4274 ± 0.0309
LCNMF	0.5849 ± 0.0209	0.7314 ± 0.0148	0.5965 ± 0.0198	0.4063 ± 0.0285	0.3672 ± 0.0298	0.4552 ± 0.0286
GOCNMF	0.6426 ± 0.0169	0.7963 ± 0.0099	0.6644 ± 0.0184	0.4900 ± 0.0262	0.4281 ± 0.0333	0.5743 ± 0.0203
SNMFWLP	0.5742 ± 0.0299	0.7667 ± 0.0144	0.6124 ± 0.0250	0.3774 ± 0.0491	0.2872 ± 0.0537	0.5640 ± 0.0176
HLPNMF	0.5935 ± 0.0175	0.7904 ± 0.0079	0.6251 ± 0.0198	0.4485 ± 0.0250	0.3829 ± 0.0351	0.5437 ± 0.0124
PCE_NMF	0.6506 ± 0.0208	0.7910 ± 0.0117	0.6671 ± 0.0197	0.4927 ± 0.0293	0.4352 ± 0.0373	0.5693 ± 0.0215

Table 8: Complete formal results on PIE. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.5380 ± 0.0167	0.7992 ± 0.0091	0.5897 ± 0.0122	0.4562 ± 0.0184	0.4129 ± 0.0222	0.5101 ± 0.0147
CNMF	0.5564 ± 0.0161	0.8020 ± 0.0082	0.6074 ± 0.0102	0.4702 ± 0.0169	0.4238 ± 0.0210	0.5283 ± 0.0121
GNMFLD	0.9416 ± 0.0061	0.9508 ± 0.0042	0.9416 ± 0.0061	0.8888 ± 0.0104	0.8826 ± 0.0119	0.8951 ± 0.0091
EDNMF	0.7682 ± 0.0140	0.8463 ± 0.0098	0.7717 ± 0.0142	0.6440 ± 0.0217	0.6196 ± 0.0218	0.6704 ± 0.0226
LCNMF	0.7565 ± 0.0095	0.8382 ± 0.0042	0.7602 ± 0.0096	0.6312 ± 0.0121	0.6126 ± 0.0149	0.6511 ± 0.0105
GOCNMF	0.9352 ± 0.0099	0.9534 ± 0.0055	0.9352 ± 0.0099	0.8829 ± 0.0147	0.8715 ± 0.0163	0.8947 ± 0.0134
SNMFWLP	0.5540 ± 0.0178	0.7549 ± 0.0170	0.6159 ± 0.0179	0.3067 ± 0.0286	0.2114 ± 0.0248	0.5622 ± 0.0255
HLPNMF	0.9589 ± 0.0088	0.9714 ± 0.0049	0.9589 ± 0.0088	0.9267 ± 0.0142	0.9186 ± 0.0168	0.9350 ± 0.0117
PCE_NMF	0.9376 ± 0.0110	0.9527 ± 0.0068	0.9376 ± 0.0110	0.8856 ± 0.0179	0.8761 ± 0.0201	0.8953 ± 0.0158

Table 9: Complete formal results on COIL100. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.4620 ± 0.0123	0.7389 ± 0.0040	0.5038 ± 0.0111	0.4114 ± 0.0084	0.3594 ± 0.0105	0.4811 ± 0.0083
CNMF	0.4776 ± 0.0178	0.7450 ± 0.0050	0.5175 ± 0.0142	0.4221 ± 0.0120	0.3714 ± 0.0146	0.4891 ± 0.0112
GNMFLD	0.8177 ± 0.0065	0.8945 ± 0.0035	0.8181 ± 0.0065	0.7413 ± 0.0087	0.7259 ± 0.0109	0.7574 ± 0.0081
EDNMF	0.5549 ± 0.0104	0.7233 ± 0.0061	0.5584 ± 0.0100	0.4366 ± 0.0100	0.3996 ± 0.0121	0.4814 ± 0.0092
LCNMF	0.5442 ± 0.0118	0.7188 ± 0.0061	0.5480 ± 0.0111	0.4288 ± 0.0102	0.3887 ± 0.0111	0.4782 ± 0.0109
GOCNMF	0.8166 ± 0.0076	0.8731 ± 0.0062	0.8166 ± 0.0076	0.6997 ± 0.0127	0.6865 ± 0.0123	0.7133 ± 0.0135
SNMFWLP	0.4131 ± 0.0178	0.6272 ± 0.0167	0.4670 ± 0.0158	0.1756 ± 0.0221	0.1161 ± 0.0180	0.3643 ± 0.0195
HLPNMF	0.8740 ± 0.0067	0.9229 ± 0.0039	0.8743 ± 0.0064	0.7961 ± 0.0100	0.7803 ± 0.0125	0.8126 ± 0.0086
PCE_NMF	0.9100 ± 0.0058	0.9452 ± 0.0030	0.9101 ± 0.0058	0.8590 ± 0.0090	0.8505 ± 0.0110	0.8677 ± 0.0072

Table 10: Complete formal results on AR. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.6094 ± 0.0192	0.8413 ± 0.0111	0.6594 ± 0.0166	0.4966 ± 0.0259	0.4326 ± 0.0271	0.5833 ± 0.0248
CNMF	0.5919 ± 0.0149	0.8327 ± 0.0082	0.6426 ± 0.0134	0.4786 ± 0.0198	0.4159 ± 0.0208	0.5640 ± 0.0216
GNMFLD	0.6142 ± 0.0137	0.8242 ± 0.0049	0.6279 ± 0.0126	0.4763 ± 0.0152	0.4240 ± 0.0204	0.5437 ± 0.0084
EDNMF	0.6355 ± 0.0220	0.8076 ± 0.0146	0.6393 ± 0.0222	0.4695 ± 0.0293	0.4032 ± 0.0325	0.5628 ± 0.0238
LCNMF	0.6902 ± 0.0131	0.8409 ± 0.0070	0.6971 ± 0.0135	0.5505 ± 0.0164	0.5106 ± 0.0184	0.5974 ± 0.0174
GOCNMF	0.6323 ± 0.0097	0.8394 ± 0.0038	0.6485 ± 0.0086	0.4889 ± 0.0134	0.4249 ± 0.0186	0.5762 ± 0.0083
SNMFWLP	0.5097 ± 0.0136	0.7761 ± 0.0092	0.5353 ± 0.0135	0.2742 ± 0.0320	0.1896 ± 0.0284	0.5040 ± 0.0183
HLPNMF	0.6390 ± 0.0148	0.8418 ± 0.0056	0.6515 ± 0.0136	0.5073 ± 0.0172	0.4494 ± 0.0218	0.5828 ± 0.0136
PCE_NMF	0.6332 ± 0.0098	0.8381 ± 0.0041	0.6481 ± 0.0091	0.4892 ± 0.0148	0.4245 ± 0.0200	0.5776 ± 0.0084

Table 11: Complete formal results on MSRA25. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.4906 ± 0.0183	0.5455 ± 0.0161	0.5084 ± 0.0153	0.3777 ± 0.0137	0.3542 ± 0.0146	0.4048 ± 0.0160
CNMF	0.5125 ± 0.0291	0.5648 ± 0.0246	0.5293 ± 0.0229	0.4000 ± 0.0237	0.3785 ± 0.0242	0.4241 ± 0.0243
GNMFLD	0.9904 ± 0.0029	0.9814 ± 0.0048	0.9904 ± 0.0029	0.9813 ± 0.0058	0.9807 ± 0.0052	0.9818 ± 0.0064
EDNMF	0.5844 ± 0.0530	0.5337 ± 0.0361	0.5942 ± 0.0483	0.4352 ± 0.0461	0.4218 ± 0.0471	0.4501 ± 0.0479
LCNMF	0.5434 ± 0.0327	0.4991 ± 0.0234	0.5529 ± 0.0292	0.3752 ± 0.0265	0.3493 ± 0.0319	0.4061 ± 0.0224
GOCNMF	0.9748 ± 0.0100	0.9588 ± 0.0133	0.9748 ± 0.0100	0.9514 ± 0.0192	0.9494 ± 0.0198	0.9533 ± 0.0188
SNMFWLP	0.9269 ± 0.0451	0.9630 ± 0.0151	0.9450 ± 0.0323	0.9283 ± 0.0397	0.8963 ± 0.0617	0.9641 ± 0.0189
HLPNMF	0.9941 ± 0.0014	0.9878 ± 0.0027	0.9941 ± 0.0014	0.9881 ± 0.0028	0.9875 ± 0.0030	0.9887 ± 0.0028
PCE_NMF	0.9788 ± 0.0082	0.9607 ± 0.0126	0.9788 ± 0.0082	0.9593 ± 0.0154	0.9595 ± 0.0142	0.9592 ± 0.0166

Table 12: Complete formal results on MNIST. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.4851 ± 0.0262	0.4524 ± 0.0144	0.5341 ± 0.0228	0.3876 ± 0.0169	0.3720 ± 0.0186	0.4047 ± 0.0163
CNMF	0.5069 ± 0.0281	0.4683 ± 0.0182	0.5588 ± 0.0252	0.4083 ± 0.0217	0.3916 ± 0.0233	0.4269 ± 0.0223
GNMFLD	0.9324 ± 0.0033	0.8528 ± 0.0047	0.9324 ± 0.0033	0.8729 ± 0.0057	0.8700 ± 0.0062	0.8758 ± 0.0051
EDNMF	0.5557 ± 0.0181	0.4577 ± 0.0092	0.5767 ± 0.0173	0.4132 ± 0.0112	0.4098 ± 0.0114	0.4167 ± 0.0111
LCNMF	0.5400 ± 0.0223	0.4490 ± 0.0103	0.5636 ± 0.0175	0.4064 ± 0.0148	0.4017 ± 0.0153	0.4113 ± 0.0147
GOCNMF	0.9352 ± 0.0027	0.8571 ± 0.0041	0.9352 ± 0.0027	0.8778 ± 0.0047	0.8755 ± 0.0051	0.8802 ± 0.0043
SNMFWLP	0.9342 ± 0.0026	0.8576 ± 0.0038	0.9342 ± 0.0026	0.8759 ± 0.0046	0.8734 ± 0.0051	0.8784 ± 0.0042
HLPNMF	0.9291 ± 0.0040	0.8508 ± 0.0046	0.9291 ± 0.0040	0.8677 ± 0.0066	0.8652 ± 0.0074	0.8702 ± 0.0058
PCE_NMF	0.9315 ± 0.0034	0.8501 ± 0.0052	0.9315 ± 0.0034	0.8711 ± 0.0058	0.8685 ± 0.0062	0.8736 ± 0.0054

Table 13: Complete formal results on USPS. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

Method	ACC	NMI	Purity	F-score	Precision	Recall
NMF	0.6671 ± 0.0305	0.6092 ± 0.0196	0.7142 ± 0.0264	0.5790 ± 0.0227	0.5874 ± 0.0245	0.5710 ± 0.0214
CNMF	0.6736 ± 0.0423	0.6108 ± 0.0188	0.7139 ± 0.0240	0.5806 ± 0.0253	0.5884 ± 0.0260	0.5731 ± 0.0252
GNMFLD	0.9685 ± 0.0010	0.9202 ± 0.0020	0.9685 ± 0.0010	0.9440 ± 0.0017	0.9389 ± 0.0019	0.9491 ± 0.0016
EDNMF	0.7713 ± 0.0022	0.6606 ± 0.0041	0.7713 ± 0.0022	0.6515 ± 0.0035	0.6523 ± 0.0045	0.6507 ± 0.0030
LCNMF	0.7652 ± 0.0192	0.6584 ± 0.0117	0.7677 ± 0.0093	0.6480 ± 0.0136	0.6527 ± 0.0126	0.6434 ± 0.0148
GOCNMF	0.9695 ± 0.0012	0.9222 ± 0.0024	0.9695 ± 0.0012	0.9459 ± 0.0021	0.9414 ± 0.0022	0.9504 ± 0.0020
SNMFWLP	0.9700 ± 0.0012	0.9235 ± 0.0023	0.9700 ± 0.0012	0.9473 ± 0.0020	0.9438 ± 0.0021	0.9507 ± 0.0020
HLPNMF	0.9719 ± 0.0011	0.9268 ± 0.0026	0.9719 ± 0.0011	0.9503 ± 0.0020	0.9474 ± 0.0022	0.9531 ± 0.0019
PCE_NMF	0.9700 ± 0.0011	0.9226 ± 0.0024	0.9700 ± 0.0011	0.9465 ± 0.0021	0.9424 ± 0.0024	0.9506 ± 0.0020

the multi-method comparison and the paired Wilcoxon test [35] with Holm adjustment [36] for the selected pairwise comparisons. The mean PCE-NMF minus HLPNMF ACC difference was 0.0031 (four wins in ten data sets; Holm-adjusted Wilcoxon $p = 0.9219$). Against GOCNMF, the difference was 0.0150 (seven wins; adjusted $p = 0.4648$). The corresponding NMI differences were -0.0050 and 0.0121, with adjusted $p = 0.5508$ and $p = 0.8457$. The lead changes across data sets: PCE-NMF leads the three-method ACC comparison on three, HLPNMF on six, and GOCNMF on one. Figure 3 summarizes the

nine methods' average ranks across the six metrics.

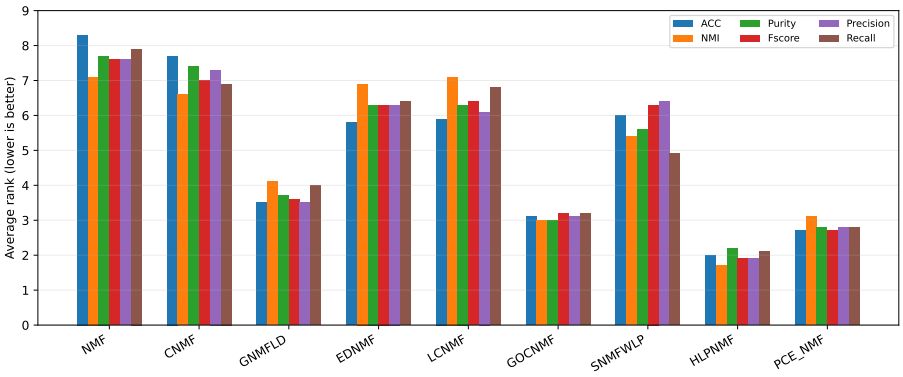


Figure 3: Average ranks of nine methods across ten data sets for six unlabeled-sample clustering metrics. Within each data set, ranking uses the method mean over 20 seeds; lower average ranks are better.

5.4. Controlled mechanism comparison

We compared PCE-NMF with HCE-NMF, which retains NMF and the labeled support constraint but removes propagated exclusion, and with a harmonic-propagation-only predictor using the same graph field. PCE-NMF exceeded HCE-NMF in ACC on all ten data sets. The mean data-set-level difference was 0.4319 (95% confidence interval 0.3124–0.5514; Holm-adjusted $p = 0.0039$). This comparison measures the gain from inserting propagated exclusion into a hard-support factorization. Against harmonic propagation alone, the mean difference was −0.0006 (95% confidence interval −0.0134–0.0121; adjusted $p = 0.6953$). The harmonic field is a strong predictor in its own right; the factor-based decision changes its performance by data set rather than yielding a consistent aggregate gain.

Table 14 resolves the aggregate comparison by data set. The factor-based rule exceeds harmonic prediction on COIL100 by 0.0243 and on COIL20 by 0.0143, whereas harmonic prediction exceeds it on YaleB by 0.0428 and on PIE by 0.0113. The same propagated field can thus support different decisions after class-aware reconstruction.

5.5. Parameter sensitivity and labeled-sample proportion

PCE-NMF parameters were selected on five development splits separate from the formal evaluation. The exclusion weight was assessed before the

Table 14: Mean unlabeled-sample ACC over 20 paired seeds in the mechanism controls. HCE-NMF uses hard labeled support without propagated exclusion; harmonic-only predicts directly from the same graph field used by PCE-NMF.

Data set	PCE-NMF	HCE-NMF	Harmonic-only
YaleB	0.6934	0.1469	0.7362
COIL20	0.9713	0.6606	0.9570
Optdigits	0.9832	0.7041	0.9825
ORL	0.6506	0.3011	0.6418
PIE	0.9376	0.3292	0.9489
COIL100	0.9100	0.3795	0.8857
AR	0.6332	0.2726	0.6297
MSRA25	0.9788	0.2298	0.9783
MNIST	0.9315	0.5771	0.9353
USPS	0.9700	0.7396	0.9702

graph-neighborhood size; data sets without an inner validation sample per class used the predeclared transfer rule. Figure 4 shows the recorded validation ACC for four representative data sets. The orange points mark locked choices; the graph-size sweep uses each data set's selected exclusion weight.

We then evaluated 5%, 10%, 15%, 20%, and 30% labeled samples per class on PIE, MSRA25, COIL20, and MNIST. The 10% points reuse the main-comparison trials. At 5% labels, PCE-NMF obtained ACC 0.9459 on COIL20 versus 0.8101 for HLPNMF. On PIE and MSRA25, HLPNMF remained higher at the same ratio (0.9094 versus 0.8854, and 0.9770 versus 0.9460). Figure 5 shows the complete preselected range.

5.6. Noise robustness

Incorrect observed labels.. We also changed 0–3 observed labels per class on COIL20 and PIE while keeping the labeled indices, data, initialization, and evaluation labels fixed within each seed. All nine methods received the same altered labels. At three incorrect labels per class, the mean paired ACC changes from the clean condition were -0.3765 for PCE-NMF, -0.3602 for HLPNMF, and -0.3812 for GOCNMF on COIL20; on PIE they were -0.5558, -0.5524, and -0.5647, respectively. The three methods show similar degradation at this corruption level. Figure 6 shows all preselected levels.

We corrupted 10%–50% of samples using either salt-and-pepper noise or a random 5×5 black occlusion, with the clean and corrupted trials

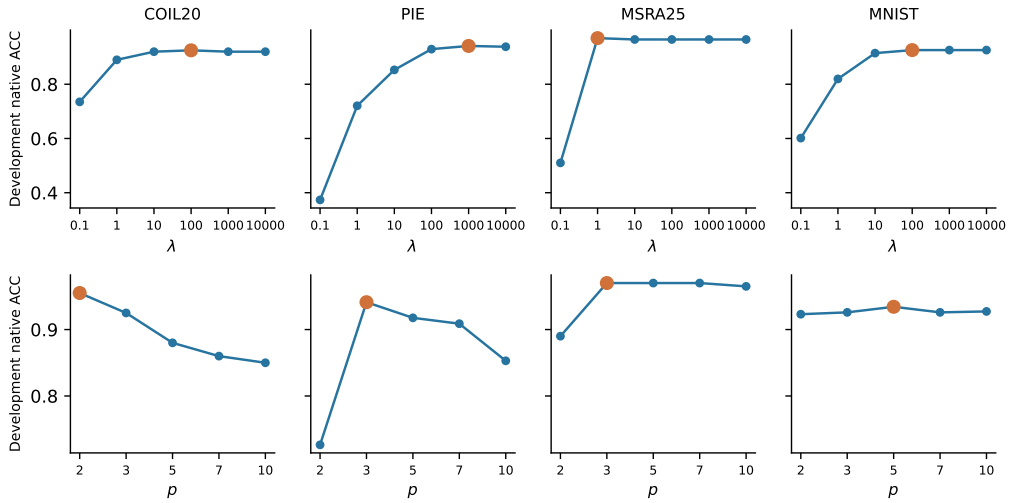


Figure 4: PCE-NMF parameter development on COIL20, PIE, MSRA25, and MNIST. Mean native validation ACC over five development splits. Orange points mark locked parameters. The upper row varies exclusion weight λ; the lower row varies graph neighborhood size p at the selected λ.

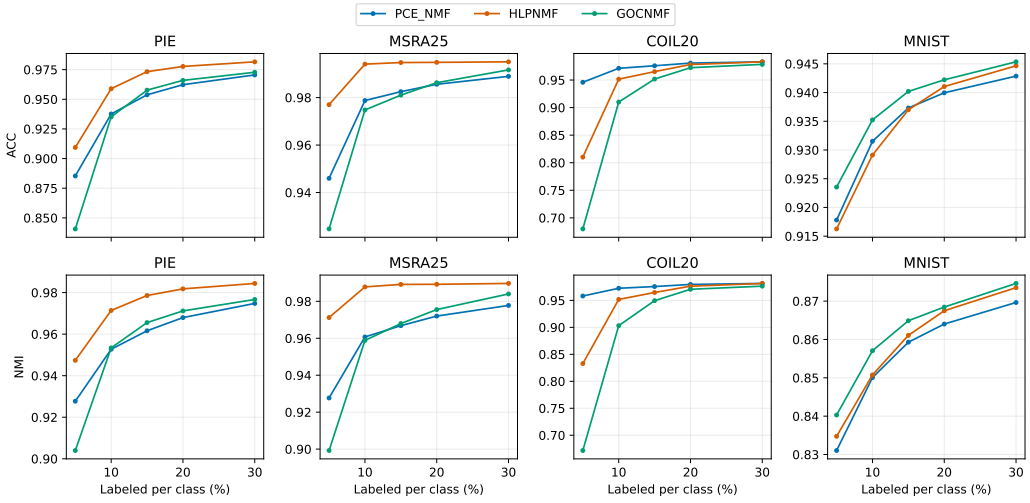


Figure 5: Mean ACC and NMI over 20 seeds at five labeled-sample proportions on four preselected data sets. The 10% values are the audited main-comparison results.

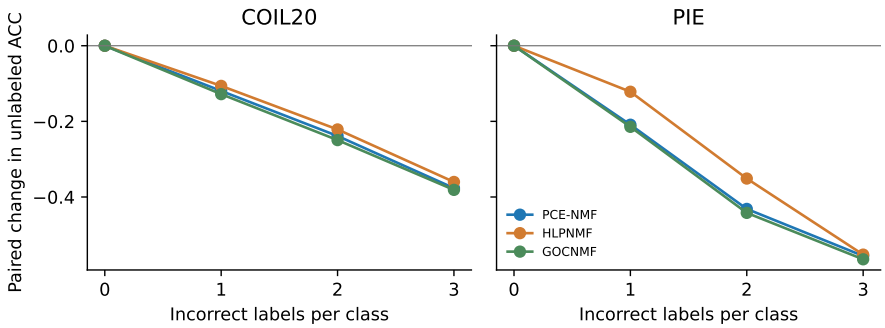


Figure 6: Mean paired change in unlabeled-sample ACC under 0–3 incorrect observed labels per class on COIL20 and PIE. Each point uses 20 matched seeds; the same altered labels are supplied to all methods within each condition.

sharing labeled splits. Figure 7 illustrates the two pixel-level perturbations on representative samples from the three image data sets.

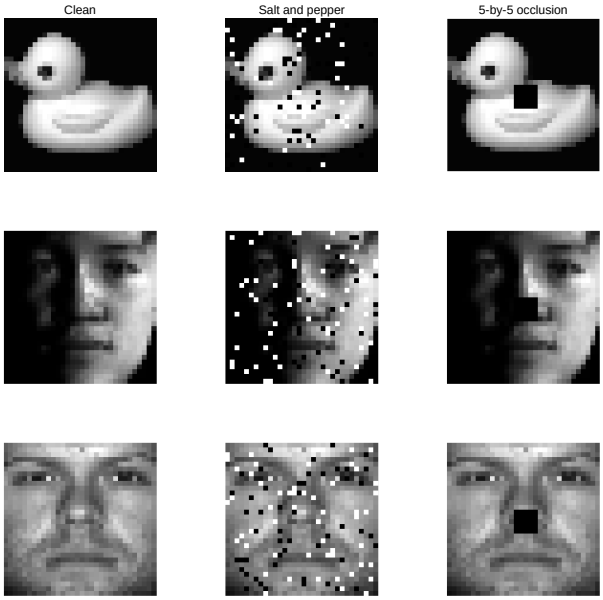


Figure 7: Representative 32-by-32 samples from COIL20, PIE, and YaleB before and after salt-and-pepper corruption (pixel probability 0.1) or a 5-by-5 black occlusion. Each row uses a common grayscale range. The images show the applied input perturbations.

Across the three selected data sets and five corruption levels, the descriptive

mean ACC changes under salt-and-pepper noise were −0.0497 for PCE-NMF, −0.0685 for HLPNMF, and −0.0414 for GOCNMF. Under occlusion, they were −0.0146, −0.0107, and −0.0061, respectively. The relative degradation depends on the corruption type and data set; Figure 8 displays the paired changes by condition.

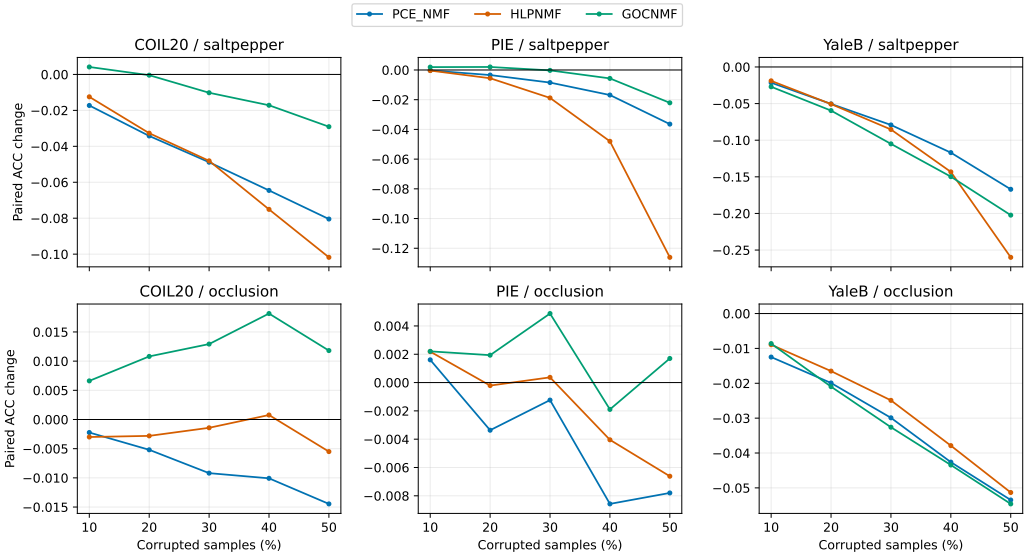


Figure 8: Mean paired change in unlabeled-sample ACC from the clean trial under salt-and-pepper noise and 5×5 occlusion. Each point uses 20 matched seeds. Columns identify the data set and rows the corruption type.

5.7. Scalability and running time

On balanced COIL100 subsets containing 20, 40, 60, 80, and 100 classes, PCE-NMF had the highest mean ACC among PCE-NMF, HLPNMF, and GOCNMF at every tested size. At 100 classes, their ACC means were 0.9100, 0.8740, and 0.8166. The corresponding measured wall times were 14.274, 123.278, and 13.254 seconds, respectively. Figure 9 shows performance and measured time as the selected COIL100 subset grows.

In a separate ten-data-set timing study, each trial rebuilt its graph and timed construction, fitting, and native prediction. PCE-NMF was faster than HLPNMF and slower than GOCNMF on every data set. On COIL100, their respective mean times were 10.752, 92.622, and 9.485 seconds. Figure 10 reports all nine methods on the same machine and implementation.

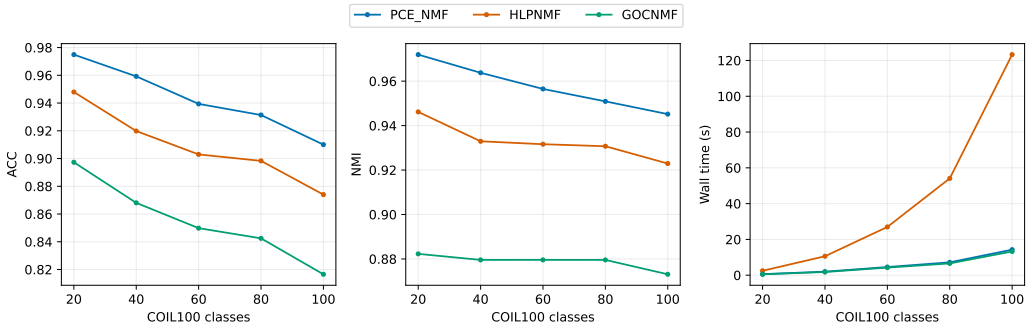


Figure 9: ACC, NMI, and measured wall time on balanced COIL100 subsets containing 20–100 classes. Each point averages 20 seeds.

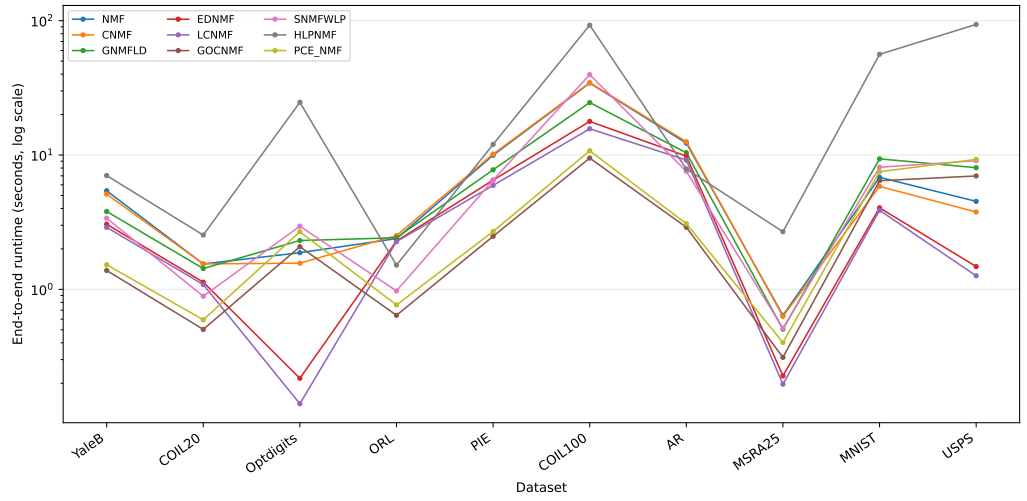


Figure 10: Mean end-to-end runtime over 20 seeds for nine methods on ten data sets, including each run's graph construction, fitting, and native prediction. No persistent graph cache was shared across timed trials. The vertical axis is logarithmic.

5.8. Numerical convergence

We recorded the original PCE-NMF objective at initialization and after each of its 50 updates in one formal run on YaleB, COIL20, PIE, and COIL100. The objective decreased at every recorded step on all four data sets. All values remained finite, and the measured nonnegativity and labeled-support violations were zero. Figure 11 shows these objective trajectories.

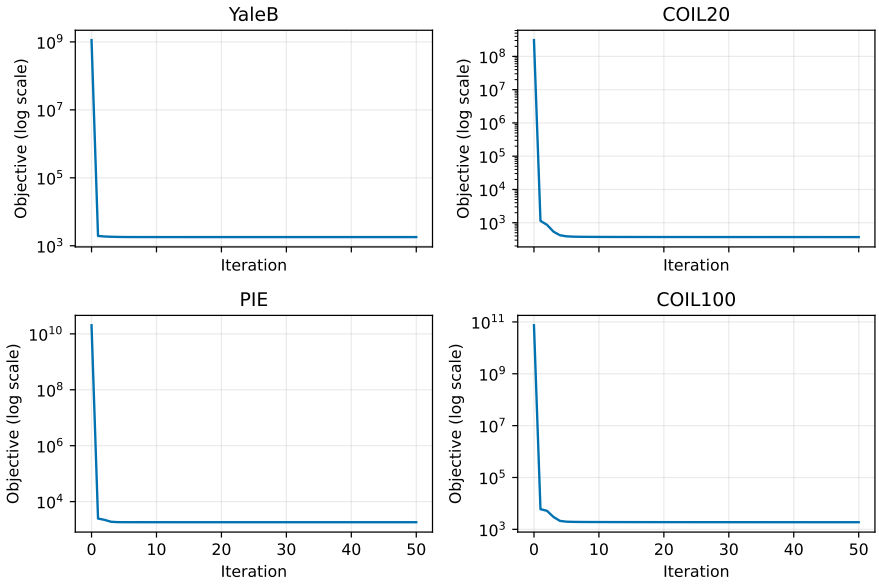


Figure 11: PCE-NMF objective trajectories over 50 updates on four representative data sets using one formal run and locked parameters. The objective axis is logarithmic.

6. Discussion

The controls identify the roles of propagation and factorization. Relative to HCE-NMF, PCE-NMF uses the same class support and NMF reconstruction while adding a class-specific penalty derived from the graph. Its gain on all ten data sets shows that this propagated preference changes the fitted factors substantially. The harmonic field also supports a direct predictor. PCE-NMF gains over that predictor on COIL20 and COIL100, while the harmonic decision leads on YaleB and PIE. The aggregate comparison is therefore close, and the factorization's predictive effect is visible at the data-set level.

The model joins a non-negative class-aware representation, explicit reconstruction, and a direct decision on scale-invariant class contributions. This combination is distinct from graph regularization of the representation and from high-order propagation through a coefficient matrix. The ten-data-set comparison shows competitive accuracy on object data, particularly COIL20 and COIL100, alongside HLPNMF's stronger results on six other data sets. In the no-cache timing study, PCE-NMF is faster than HLPNMF and slower than GOCNMF on every data set. Its graph construction requires quadratic pairwise distance work before neighbor selection; the reported times include that cost. The class-count experiment describes the tested balanced COIL100 subsets.

7. Conclusion

PCE-NMF incorporates harmonic class evidence into the energy of class-specific NMF contributions and uses the same contributions for direct prediction. Its class score is invariant to factor rescaling, and the exact alternating updates decrease the objective under the stated conditions. Propagated exclusion improved accuracy over matched hard-support factorization on all ten data sets. The factor decision led the harmonic predictor on COIL20 and COIL100 and followed it on YaleB and PIE. In the nine-method comparison, PCE-NMF achieved the highest ACC among the three closely related methods on COIL20, ORL, and COIL100. Together, these results establish a class-aware factorization whose optimization and prediction use a common, graph-guided contribution measure.

CRediT authorship contribution statement

Deshu Sun: Writing-original draft, Methodology, Software, Conceptualization, Formal analysis. Feng Wang: Writing-review and editing, Funding acquisition, Visualization, Conceptualization. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The processed matrices, MATLAB R2019a experiment drivers, locked parameters, raw results, and figure source data are available at `https://github.com/matrix-research-lab-Wf/PCE-NMF`. The external HLPNMF author implementation is obtained from its original source as documented in the repository.

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